ZS-Q17
The Self-Mediation No-Go and the Reach Bound of Z-Spin Quantum Transport: A Complete Formalization, with the Spontaneous-Radiation and Fifth-Force Gates Closed

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Verification: reach-bound suite 14/14 PASS | No-Go DERIVED (no-programming + Lawvere, PBT-completed) | radiation gate F-Q1-rad CLOSED (structural + parametric) | 9-step protocol PASS | Zero Free Parameters | Anti-Numerology: N/A (the reach bound is a derived physical scale; the No-Go and gate closures are structural)

§0. Abstract

We replace the v1.x question — *can matter travel by becoming a wave?*, which resolves to the well-known sub-luminal teleportation channel and yields no new Z-Spin content — with two questions whose answers *are* Z-Spin-specific, and we follow each to a definite epistemic terminus using only the locked corpus inputs (**A** = 35/437, **Q** = 11, dim(**Z**) = 2) and imported *proven external mathematics*.

(i) The Reach Bound [DERIVED / TESTABLE]. A massive payload of mass m can be coherently teleported through the rank-2 Z-bottleneck only over a maximum distance **L_max(m) = c·τ_ent = c·ℏ/(2A·E_diff)**, set by the entangled-resource half-life of Z-Spin gravitational decoherence (ZS-Q1 $\tau_D = \hbar/(A\cdot E_{diff})$; ZS-Q2 $\tau_{ent} = \tau_{single}/2$). The bound scales as **L_max ∝ m^(−5/3)** at fixed density and reproduces both heritage anchors (gold 10^9 amu → $\tau_{single} \approx 7$ days; $M_{crit}(\tau = 1\ s) \approx 2.0\times10^{12}$ amu). Its falsifiable signature is the mass-independent ratio **L_max(Z-Spin)/L_max(Penrose) = 1/A = 12.49**. A **Horizon Corollary** answers the singularity question: compressing a mass to its Schwarzschild radius gives **E_diff = (3/10)*mc*²** exactly (mass-independent), so the X→Y ferrying rate $\Gamma \propto m^2$ diverges with compression — the rest-energy-scale limit of the same Γ mechanism, matching ZS-A3 (the horizon is the Z-sector boundary; the interior *r*↔*t* exchange is the X↔Y sector exchange).

(ii) The Self-Mediation No-Go [DERIVED], completely formalized. No Z-Spin-mediated CPTP channel can faithfully transport a state encoding *its own mediation operation*. We establish this on two independent proven anchors — the **Nielsen–Chuang no-programming theorem** (a dim-2 register cannot deterministically program the **Q** = 11 mediation) and the **Lawvere diagonal theorem** with the uniqueness of the i-tetration fixed point z^* (a self-transport would be a second self-referential fixed point) — and we *complete* the formalization with **port-based teleportation** (Ishizaka–Hiroshima; Christandl *et al.*): the deterministic-exact statement is PROVEN, while the approximate case is governed by the fidelity ceiling **F_d(N) ≤ √N/d** (d = 2), with the port count N identified with the serial Z-handshake count and unit

fidelity reached only in the $n \to \infty$ (i-tetration) limit. The No-Go thus sharpens from "impossible" to a quantitative fidelity bound.

(iii) Unification [DERIVED-interpretation]. The No-Go is the same Lawvere obstruction as the ZS-F0 bootstrap **B0** (the framework cannot self-apply to derive its own foundation) and the ZS-Q16 strong-outcome **OPEN** (the channel supplies no rule to select among its own outcomes) — three faces of *"the mediator cannot mediate itself,"* with z^* the unique fixed point.

(iv) Experimental status; the radiation and fifth-force gates [NEW]. The discriminating direct test of (i) is the ratio-12.49 measurement, awaiting 10^9–10^{12} amu nanosphere interferometry (2028–2032); the reach bound itself maps onto gravitationally-induced-entanglement tests. We then ask whether Z-Spin's σ_z geometric decoherence is already constrained by the underground X-ray experiments that excluded parameter-free Diósi–Penrose (Donadi *et al.*; Majorana; XENONnT). We answer **no**, and close the gate **F-Q1-rad**: ZS-Q1 is a scalar-tensor / *environmental* decoherence model (rate ∝ E_diff, vanishing for unsuperposed matter; linear-Lindblad ensemble, ZS-Q16), not a fundamental stochastic-collapse model, so it carries **no universal radiating term**; and its scalar **ε** is a *massive Yukawa mediator* that **decouples** when heavy — the opposite of the Diósi–Penrose small-smearing $1/R_0^3$ enhancement, with any residual ε-vacuum effect **$(A \cdot G)^2$-suppressed**. A separate gate **F-Q1-5th** is registered: the A = 0.08 non-minimal coupling implies a potential fifth force, requiring ε to be heavy/short-range or screened (the Planck-scale Z-sector reading satisfies this).

The v1.x faithful-sub-luminal-teleportation and no-signaling results are retained as **Tier-1 consistency** (standard quantum information; no Z-Spin prediction), per the ZS-Q2 tiering discipline.

Epistemic Status Legend

STATUS	DEFINITION
PROVEN	Mathematical theorem from (Z,X,Y)=(2,3,6) and/or imported proven external mathematics; falsifiable only by logical error.
DERIVED	Consequence of PROVEN items plus the Z-Spin action, zero free parameters beyond A; falsifiable by experiment.
DERIVED-CONDITIONAL	Derived conditional on an explicitly stated assumption tracked in the paper.
DERIVED-interpretation	Synthetic reading combining PROVEN/DERIVED corpus theorems without new axioms.
TESTABLE	Pre-registered prediction with an explicit falsification protocol and timeline.

STATUS	DEFINITION
NON-CLAIM	Explicit declaration of what is NOT asserted; bounds the framework's scope.
OPEN	Recognized gap honestly registered; promotion path may or may not exist.
BOOTSTRAP-HYPOTHESIS	Meta-logical founding axiom (B0); not derivable within the framework.
LOCKED	Core constant fixed upstream; not modified here (A = 35/437, Q = 11, dim Z = 2).

§1. Introduction

The intuition that motivated v1.0 — convert matter to a delocalized Y-sector wave, exploit the wave's global character, and re-collapse at a chosen remote X-location — maps inside Z-Spin onto the measurement/decoherence cycle itself (ZS-Q1). v1.x established, correctly, that as a *controllable transport channel* this is the standard teleportation protocol (Bennett *et al.* 1993): faithful but sub-luminal, with the super-luminal variant blocked by Gisin–Polchinski. These conclusions are sound but are re-labellings of textbook quantum information; they introduce no new mathematical structure, no new derived scale, and no new falsifiable prediction. We therefore demote them to Tier-1 consistency (§8) and ask instead the two Z-Spin-specific questions.

First (the positive side): how far can a given payload be carried before the framework's own gravitational decoherence destroys the resource? This has a parameter-free answer, the **Reach Bound** (§3), with a falsifiable 1/A signature and a clean strong-gravity limit that answers the black-hole singularity question. Second (the structural side): the resource, channel, and reconstruction legs of a Z-Spin teleport are themselves Z-bottleneck handshakes (the Bell measurement is a rank-2 Z-mediated CPTP map, ZS-Q1 §3.3). What if the payload *is* the mediation? We prove it cannot be — the **Self-Mediation No-Go** (§5) — and we complete its formalization with port-based teleportation (§5.5). Finally we confront the framework with current data: the reach bound's discriminating test, and whether Z-Spin's decoherence is already constrained by the X-ray experiments that killed parameter-free Diósi–Penrose (§7).

§2. Locked Inputs

All quantities are inherited from prior corpus papers and the cited external theorems. No new constant or free parameter is introduced.

Table 1. Locked and imported inputs to ZS-Q17 v2.1.

Quantity	Value / Statement	Source	Status
A (geometric impedance)	$35/437 = 0.0800915$	ZS-F2 v1.0	**LOCKED**
(Z, X, Y); **Q**	(2, 3, 6); 11	ZS-F5 v1.0	**PROVEN**
$L_{XY} \equiv 0$ (X–Y vanishing)	exact zero (Z-Spin mediation forced)	ZS-F1 v1.0	**PROVEN**
dim(Z) = 2 Kraus rank	Stinespring dilation (CPTP)	ZS-Q1 v1.0 §3.3	**PROVEN**
Z-channel capacity ≤ ln 2	$\mathrm{rank}(T_{XY}) \leq \dim(Z) = 2$	ZS-Q7 v1.0 Thm 2	**DERIVED**
$\tau_D = \hbar/(A \cdot E_{diff})$	geometric decoherence time	ZS-Q1 v1.0 §5.1	**DERIVED**
$E_{diff} = (3/5)G\, m^2/R$	Newtonian self-energy (sphere)	ZS-Q1 v1.0 §5.1	**DERIVED**
$\tau_D/\tau_{Penrose} = 1/A = 12.49$	parameter-free signature	ZS-Q1 v1.0	**DERIVED**
$\tau_{ent} = \tau_{single}/2$	entangled-pair half-life	ZS-Q2 v1.0 §7.3	**DERIVED**
Lindblad operator σ_z; linear ensemble	Hermitian; ensemble obeys linear Lindblad	ZS-Q1 §3.4; ZS-Q16 §3	**DERIVED**
Action $S = \int\sqrt{(-g)}[(1+A\varepsilon^2)R/2 - (\partial\varepsilon)^2/2 - V(\varepsilon)]$	ε massive dynamical scalar; attractor $\varepsilon=1$; U(1) Z-bias Φ, vortex core $\lvert\Phi\rvert=0$	ZS-F1 v1.0	**LOCKED**
i-tetration fixed point z^*	$z^* = -W_0(-i\pi/2)/(i\pi/2)$, unique	ZS-M1 v1.0 (HSI)	**PROVEN**
Lawvere diagonal (B0→B1)	self-reference ⇒ fixed point	ZS-F0 v1.0(R) §2.2, §11	**DERIVED** (imported)
No-programming theorem	dim(program) ≥ #distinct unitaries	Nielsen–Chuang 1997	**PROVEN (ext.)**
Port-based teleportation bound	$F_d(N) \leq \sqrt{N}/d$ ($N \leq d^2/2$)	Christandl *et al.* 2021	**PROVEN (ext.)**
Strong-outcome selection	dynamical single-outcome rule	ZS-Q16 v2.5 §7	**OPEN**
Horizon = Z-boundary; $r \leftrightarrow t$ = X↔Y	interior sector exchange	ZS-A3 v1.0 §(interior)	**HYPOTHESIS**

§3. The Reach Bound of Z-Spin Transport [DERIVED / TESTABLE]

§3.1 Derivation. A teleport requires a pre-shared entangled pair; reconstruction at P2 cannot complete until the classical record arrives at $v \le c$. The protocol succeeds only if the entangled resource survives at least the light-travel time *L/c*. Each Bell-pair half decoheres under Z-Spin gravitational dephasing ($\Gamma = 2A(\Delta E/\hbar)^2$ at the attractor; ZS-Q1 §3.4); the concurrence decays at 2Γ, giving the entangled half-life **τ_ent = τ_single/2 = ℏ/(2A·E_diff)** (ZS-Q2 §7.3), with E_diff = (3/5)G m^2/R. Requiring τ_ent ≥ L/c yields

$$\mathbf{L\_max(m) = c \cdot \tau\_ent = c \cdot \hbar / (2A \cdot E\_diff) = (5\, c\, \hbar\, R) / (6\, A\, G\, m^2)} \quad (1)$$

For fixed density ρ, R = $(3m/4\pi\rho)^{(1/3)}$, so **L_max ∝ m^(−5/3)** — verified to eight digits (companion script Category F).

§3.2 The reach table and the 1/A signature.

Table 2. Reach bound for gold-density payloads (self-consistent R).

Payload	**Mass (amu)**	**R**	**τ_single**	**L_max**
C_{60}-scale	10^3	0.27 nm	1.0×10^8 yr	5.2×10^7 ly
Large virus	10^6	2.7 nm	1.0×10^3 yr	5.2×10^2 ly
Gold nanosphere	10^9	27 nm	3.8 days	3.3×10^2 AU
M_crit	2×10^{12}	0.35 μm	1.0 s	1.5×10^5 km
Schrödinger cat	10^{34}	5.9 m	7.0×10^{-37} s	1.0×10^{-28} m

For light payloads the reach is cosmological (gravitational decoherence irrelevant — consistent with routine photonic teleportation). At and beyond the critical mass the reach collapses below the payload's own size (the cat has L_max ≈ 10^{-28} m, twenty-nine orders below its body radius), the reach-bound form of the total decoherence that ZS-Q2 §9.4 expresses as a 10^{-34} area-law deficit. The falsifiable content is the heritage signature in spatial form,

L_max(Z-Spin) / L_max(Penrose–Diósi) = 1/A = 12.49 (2, mass-independent, DERIVED)

verified to 3.6×10^{-15} across 25 masses. [STATUS: bound DERIVED; the discriminating measurement TESTABLE, inheriting the ZS-Q1 gate (2028–2032), see §7.1.]

§3.3 Horizon Corollary — the singularity limit [DERIVED]. Compress a mass to its Schwarzschild radius R_s = 2Gm/c^2. Then

$$E_{diff} = (3/5)G\, m^2/R_s = (3/10)\, m c^2 \quad (3)$$

a fixed fraction of the rest energy, independent of mass (verified to 1.1×10^{-16}). The single-system coherence time at the horizon is $\tau_H = 10\hbar/(3A\, mc^2)$ and the X→Y ferrying rate $\Gamma = 2A((3/10)mc^2/\hbar)^2 \propto m^2$ diverges with compression — the E_{diff} → rest-energy-scale limit of the same $\Gamma = 2A(\Delta E/\hbar)^2$ mechanism. ZS-A3 v1.0 records the geometric counterpart: inside the horizon r is timelike and t spacelike (an $r\leftrightarrow t$ exchange identified with X↔Y), the horizon is the Z-sector boundary ($\varepsilon = 0$), and a black hole is "an X-sector structure collapsed into a Y-sector prison." The singularity is where coherent X-transport reach vanishes and the Z-Spin phase-rotation ferries X-sector (space + particle) information into the Y-sector at the maximal, rest-energy-set rate. Full horizon thermodynamics remain with ZS-A3/A4/A6; this corollary asserts only the Newtonian $(3/10)mc^2$ rate identity (DERIVED) and the inherited $r\leftrightarrow t$ = X↔Y reading (HYPOTHESIS).

§4. Five-Step Deep Exploration

§4.1 Step 0 — Long List (7) and Step 1 — Issue List (3, MECE). Long list: (1) fidelity deviation from A; (2) maximum coherent transport reach; (3) a self-teleportation NO-GO; (4) handshake count as i-tetration iteration with $|\lambda|$ as a transport observable; (5) a fundamental bit-rate ceiling; (6) the dual-clock proper-time compression as a near-term observable; (7) entanglement-assisted transport as the area-law origin. **Dropped:** (6) not near-term testable (over-narrated in v1.x); (7) overlaps the registered ZS-Q2 Tier-3 conjecture; (5) absorbed into (4); (1) absorbed into (2) (an ideal rank-2 channel is exactly faithful, so A-dependence enters only through decoherence). **Issue list (influence order):** I1 reach bound → I2 self-mediation no-go → I3 i-tetration transport dynamics.

§4.2 Step 2 — Issue Tree and Step 3 — Traversal.

Table 3. Issue-tree traversal with epistemic status.

Node	Question	Status	Finding
I1a	Reach bound f(A, m)	**DERIVED**	Eq. (1); composition of ZS-Q1 τ_D and ZS-Q2 τ_{ent}.
I1b	Falsifiable 1/A signature	**TESTABLE**	Eq. (2); inherits the ZS-Q1 nanosphere gate.
I2a	Self-state transport possible?	**DERIVED**	NO — no-programming: dim(Z)=2 cannot program the Q=11 mediation (§5.2).

Node	Question	Status	Finding
I2b	Lawvere / z^* connection	**DERIVED**	NO — a self-transport is a second self-referential fixed point; z^* is unique (§5.3).
I3a	Transport = i-tetration iteration?	**DERIVED**	YES — n is the handshake clock; $(R \circ E)^n \to z^*$ (ZS-F0/F10); completed in §5.5.
I3b	$\lvert\lambda\rvert=0.89151$ as fidelity decay?	**NON-CLAIM**	$\lvert\lambda\rvert$ is the self-referential-map contraction, not a fidelity; conflation is the numerology ZS-Q12 warns against.

§4.3 Step 4 — Convergence and Step 5 — Scoring. Cycle 1 settles all six nodes; cycle 2 changes zero. State-change count 6 → 0 (monotone decreasing) — the $|f'(z^*)| < 1$ analogue: **CONVERGENCE**. **Scoring:** I2 (the No-Go) is the highest-value node — structurally novel, of the NO-GO type the corpus values, and elevatable from HYPOTHESIS to DERIVED by imported proven mathematics (executed §5) and *completable* (§5.5). I1 is DERIVED and falsifiable; I3a frames the dynamics; I3b is the boundary the framework must not cross.

§5. The Self-Mediation No-Go [DERIVED]

§5.1 Statement. Theorem Q17.1 (Z-Self-Mediation No-Go). Let **Λ_Z** be a Z-Spin-mediated CPTP channel realized as the rank-2 Z-bottleneck handshake of ZS-Q1 (Stinespring dilation on $\mathcal{H}\_X \otimes \mathcal{H}\_Z$, dim(**Z**) = 2, with $L\_XY \equiv 0$ forcing this single conduit). There is no Z-Spin-mediated transport protocol that faithfully relocates a state encoding **Λ_Z** itself — that carries, through the Z-bottleneck, the data specifying the mediation operation. The mediator can transport the X-states (matter) and Y-states (waves) it mediates; it cannot transport itself. Two independent routes establish this; over-determination by routes sharing no premise beyond the locked inputs registers **DERIVED** (the corpus's DERIVED-CONDITIONAL-strong → DERIVED grading, ZS-F10).

§5.2 Route 1 — the No-Programming Obstruction (rigorous). A protocol relocating **Λ_Z** would, at the receiving locus, *reconstruct* it from transmitted data — a programmable channel with a program register specifying which operation to apply. *Imported PROVEN — Nielsen & Chuang (1997):* for a fixed gate array G with $G(|\psi\rangle\_data \otimes |P\_U\rangle\_prog) = (U|\psi\rangle)\_data \otimes |P'\_U\rangle\_prog$ for all $|\psi\rangle$, distinct $U \neq V$ require $\langle P\_U|P\_V\rangle = 0$; hence dim(program) ≥ #distinct unitaries. *Z-Spin reinterpretation:* **Λ_Z** ranges over $\gg 2$ inequivalent cross-sector operations on the **Q** = 11 register, but the only conduit is the Z-register, dim(**Z**) = 2 (capacity ≤ ln 2, ZS-Q7). Since 2 < the number to be encoded, the bound is violated — a dim-2 register cannot program the dim-$\gg$2 operation it would carry. The faithful transport of a single external qubit is the $N = 2$ special case and is permitted (the v1.x faithful-teleportation content, §8); the self-referential payload is forbidden.

§5.3 Route 2 — the Lawvere Diagonal and z^* Uniqueness (structural over-determination). A faithful self-transport would be a fixed point of the self-application endomap on the operation space. *Imported PROVEN — Lawvere (1969), FOUNDATIONAL in ZS-F0 §2.2, §11:* point-surjectivity $A \to Y^A$ forces a fixed point for every endomap of Y. *Z-Spin reinterpretation:* the register's self-referential dynamics is $T(z) = i^z$ (ZS-M1 HSI), whose **unique** attractor is $z^* = 0.4382829 + 0.3605925i$ (Lambert W $k = 0$ uniqueness), already the collapse attractor (ZS-Q12/Q16). A self-transport would be a *second* self-referential fixed point; uniqueness forbids it**.** Wootters–Zurek no-cloning and the three ZS-Q12 No-Gos are special cases of the same skeleton, so Theorem Q17.1 sits in an established lineage.

§5.4 Independence of the two routes. Route 1 is information-theoretic (program-register dimension counting); Route 2 is categorical/dynamical (fixed-point uniqueness). They share no premise beyond the locked (dim Z = 2, z^*) inputs and reach the same conclusion — over-determination registering **DERIVED**.

§5.5 Complete Formalization via Port-Based Teleportation [NEW v2.1; DERIVED]. The no-programming theorem forbids *deterministic, exact* self-programming. The complete formalization must address the *approximate* and *probabilistic* cases, and port-based teleportation (PBT) supplies them. *Imported PROVEN — Ishizaka–Hiroshima (2008); Christandl et al. (2021); Studziński et al.:* PBT is a teleportation in which the receiver applies no correction (it selects a port), and its unitary equivariance lets it act as an *approximate* (deterministic-inexact, dPBT) or *probabilistic-exact* (pPBT) universal programmable processor with *finite* resources; faithfulness is reached only as the number of ports $N \to \infty$. The deterministic entanglement fidelity obeys the closed bound

$$\mathbf{F\_d(N) \le \sqrt{N} / d \quad (for\ N \le d^2/2), \quad F\_d(N) \le 1 - (d^2-1)/(16\, N^2) \ \ otherwise} \qquad (4)$$

with exact closed forms known for d and $N \in \{2, 3, 4\}$. *Z-Spin reinterpretation.* The Z-bottleneck supplies one rank-2 channel use per transit (capacity $\le \ln 2$, ZS-Q7), i.e. $d = 2$ and one "port" per Z-handshake. A *self-mediation* attempted with N serial handshakes (the ZS-T1 macroscopic-payload reading) is then a $d = 2$ PBT with N ports, whose fidelity is ceilinged by (4): **F < 1 for any finite N**, approaching unity only as $N \to \infty$. Identifying the port count N with the stroboscopic handshake count n (ZS-F10), and recalling that the continuum limit of $(R\circ E)^n$ is the i-tetration flow toward z^* (ZS-F0 Lemma 5.2.A, ZS-M1), the No-Go acquires its **complete, quantitative form**:

> *self-mediation is impossible deterministically and exactly; approximately it is possible only with a fidelity ceilinged by $F_2(n)$ of (4), reaching unit fidelity solely in the $n \to \infty$ (i-tetration, z^*) limit.*

This closes node I3a operationally: the handshake count is the PBT port count, and the z*-convergence is the $N \to \infty$ asymptote of faithful self-programming. [STATUS: DERIVED — deterministic-exact PROVEN (no-programming); approximate ceiling PROVEN (PBT) and reinterpreted via the corpus handshake/i-tetration structure; zero free parameters. Anti-numerology: N/A.]

§6. Unification: Three Faces of One Obstruction [DERIVED-interpretation]

Table 4. Three faces of the Lawvere self-reference obstruction.

Face	Statement	Corpus status	Reading
Bootstrap B0	The framework cannot self-apply to derive its founding axiom.	**BOOTSTRAP-HYPOTHESIS** (ZS-F0)	The theory cannot transport/derive its own foundation.
Strong outcome	No dynamical rule selects one outcome among the dim Z = 2 innovation.	**OPEN**, non-epistemic (ZS-Q16 §7)	The channel cannot select among its own outcomes.
Self-mediation	The mediator cannot transport its own operation.	**DERIVED** (§5)	The mediator cannot mediate itself.

All three are the Lawvere diagonal with z^* the unique fixed point — at the meta-logical (B0), single-run dynamical (strong outcome), and channel (self-mediation) layers. This promotes neither B0 nor the strong-outcome problem; each retains its status, and the ZS-Q16 consciousness firewall (NC-Q7.4 / NC-A7.6 / NC-F10.3 / NC-F11.1 / NC-F19.1) is untouched. The contribution is to register them as one obstruction with three projections, in the spirit of ZS-F18's meta-map.

§7. Experimental Status and the Spontaneous-Radiation / Fifth-Force Gates [NEW v2.1]

§7.1 The 2026 experimental landscape. The direct, discriminating test of the Reach Bound (and of the underlying ZS-Q1 ratio 12.49) is interferometric: a measurement of the entangled-resource coherence reach (or the single-system decoherence time) for a 10^9–10^{12} amu payload, distinguishing the Z-Spin value (12.49 × Penrose) from Penrose–Diósi (ratio 1) and from tunable CSL. The current state of the field places this in the **2028–2032** window: the largest superposition over distances comparable to size remains ≈ 25 kDa (macromolecule interferometry), recently extended to metal clusters above 170 kDa; levitated nanospheres (≈ 143 nm silica) have been cooled to the motional ground state, with matter-wave interference at the billion-amu scale the stated next goal (a three-to-four-order extension). The reach bound's entangled-resource form maps directly onto **gravitationally-induced-entanglement (GIE)** experiments now in development with magnetically levitated masses (Großardt 2025 shows the Diósi–Penrose model itself predicts GIE), where the Z-Spin signature would be a 12.49 × longer GIE coherence. [STATUS of (i): bound DERIVED; discriminating test TESTABLE, 2028–2032.]

§7.2 Does Z-Spin σ_z decoherence radiate? Two model classes. The underground experiments — Donadi *et al.* (Gran Sasso Ge, 2021); the Majorana Demonstrator (2022); XENONnT (2025) — searched for the spontaneous X-ray emission predicted by collapse models and **excluded the natural parameter-free Diósi–Penrose model**, bounding it to $R_0 \gtrsim 0.5$ Å. By the Bassi–Donadi criterion the emission rate is proportional to the charged-particle **momentum-diffusion coefficient D_pp**. Crucially, these bounds apply to the **stochastic-collapse class**: a *fundamental universal white-noise field* added to the dynamics that localizes every particle continuously — even unsuperposed — giving an always-on D_pp; smaller smearing R_0 ⇒ more radiation; parameter-free $R_0 \to 0$ diverges and was excluded.

ZS-Q1 is **not** in this class. It is a **scalar-tensor / environmental** decoherence model: the scalar **ε** is a deterministic dynamical field (kinetic + V(ε), non-minimally coupled to curvature), not an ad-hoc stochastic localization term; the decoherence rate is **gated** (∝ E_diff, the branch self-energy *difference*, which vanishes for unsuperposed detector matter); and the stochasticity of the SSE is a quantum-trajectory *unraveling* whose ensemble obeys the **linear** Lindblad equation (ZS-Q16) — the defining property that there is **no universal noise**. Hence ZS-Q1 predicts no spontaneous X-rays from unsuperposed matter, exactly as Penrose's own (non-stochastic) OR does not.

§7.3 The Yukawa-mediator correction and the (A·G)² suppression. A naive worry treats the ε Compton wavelength λ_ε = ℏ/(m_ε c) as a Diósi–Penrose *smearing* length, with D_pp ∝ $1/R_0^3$ (smaller ⇒ more radiation, excluded below ≈ 0.12 Å). This is a category error. **ε is a massive Yukawa mediator**, so a *heavy* ε **decouples** (interaction ∝ e^(−r/λ_ε)) — giving *less* effect, the **opposite** of the Diósi–Penrose $1/R_0^3$ enhancement. There is no divergence to exclude. Expanding ε = 1 + δε, the linear coupling A·δε·R sources δε ∝ A·G·ρ, i.e. ε mediates a Yukawa "fifth force" of strength ~ A; the residual decoherence from δε vacuum/thermal fluctuations is therefore a **second-order gravitational effect** ∝ **(A·G)²**, with the ε mass furnishing a natural cutoff and no enhancement. For the corpus reading — ε the Planck-scale Z-sector field, λ_ε ~ ℓ_Planck — ε decouples completely from keV-scale physics. In **every** limit of m_ε the predicted radiation lies far below the exclusion.

§7.4 Gate verdicts. F-Q1-rad [CLOSED, in Z-Spin's favour]: ZS-Q1 carries no universal radiating term (scalar-tensor/environmental, gated, linear-Lindblad ensemble), and its ε mediator decouples when heavy with any residual effect (A·G)²-suppressed; Z-Spin escapes the bounds that killed parameter-free Diósi–Penrose. The closure holds *by the definition* of ZS-Q1 as a scalar-tensor model. **New consistency requirement (registered):** ZS-Q1 must remain scalar-tensor/environmental and must not be reformulated as a fundamental stochastic-collapse model with a universal noise of correlation length $R_0 \lesssim 0.12$ Å. **F-Q1-5th [NEW, registered]:** the A = 0.08 non-minimal coupling implies a potential fifth force of strength ~ A; an unscreened such Yukawa is excluded by Eöt-Wash/planetary tests over a broad range, so ε must be heavy/short-range (decoupled) or screened (chameleon/symmetron) — the Planck-scale Z-sector reading satisfies this. This is a separate gate for ZS-Q1/ZS-F1 to address explicitly.

§7.5 The two-edged sword. The very feature that lets Z-Spin escape the radiation bounds — gravitational gating, no universal noise — is the same feature that makes it untestable by radiation. The discriminating test is therefore necessarily interferometric (§7.1, ratio 12.49, 2028–2032), or GIE for the reach-bound form. [STATUS of (iv): F-Q1-rad CLOSED (structural + parametric); F-Q1-5th OPEN-registered; the radiation analysis is order-of-magnitude pending V(ε), but the structural closure does not depend on m_ε.]

§8. Retained Tier-1 Consistency (the v1.x teleportation content, demoted)

(T1) Faithful sub-luminal teleportation. An external X-sector qubit is transported P1→P2 with unit fidelity through the rank-2 Z-bottleneck given a Bell pair and a classical correction (Bennett *et al.* 1993 via ZS-Q1 CPTP). This is the $N = 2$ special case of §5.2 — consistency, not prediction. **(T2) No super-luminal relocation.** Instantaneous relocation requires a controllable ensemble-level nonlinear collapse, which by Gisin–Polchinski entails super-luminal signaling, violating F-Q2.5 and triggering F-EP.3. Standard, consistency-level. Both are filed as ZS-Q2 files Bell/CHSH/no-signaling under Tier-1; neither is counted among the new results.

§9. Falsification Gates

Gate	Condition	Type
F-Q17.1	A finite quantum channel deterministically programs an operation on a register strictly larger than itself ⇒ no-programming violated ⇒ Theorem Q17.1 Route 1 withdrawn.	**EXTERNAL**
F-Q17.2	A second attracting fixed point of $z = i^z$ is found ⇒ z^* uniqueness false ⇒ Route 2 (and the ZS-F0/M1 spine) collapses.	**MATHEMATICAL**
F-Q17.3	Entangled-pair coherence survives gravitational decoherence beyond $\tau\_ent = \hbar/(2A \cdot E\_diff)$ for a given mass ⇒ Reach Bound and 1/A signature falsified.	**OBSERVATIONAL (2028–2032)**
F-Q17.4	Nanosphere entangled-resource reach yields ratio 1.0 ± 0.5 vs Penrose ⇒ Z-Spin reach signature falsified.	**OBSERVATIONAL (2028–2032)**
F-Q17.5	A dynamical single-outcome selection rule is derived from the Z-Spin action ⇒ strong-outcome face (§6) promotes from OPEN.	**PROMOTION**
F-Q17.6	The horizon $r \leftrightarrow t$ = X↔Y identification is shown inconsistent with the ZS-A3 ε-field EFT ⇒ Horizon Corollary's interior reading withdrawn (the $(3/10)mc^2$ rate survives).	**EXTERNAL (intra-corpus)**
F-Q17.7	A deterministic, exact self-mediation channel is realized at finite resources beating the PBT ceiling $F_2(N)$ of Eq. (4) ⇒ §5.5 completion withdrawn.	**EXTERNAL**
F-Q1-rad	ZS-Q1 is shown to entail a universal always-on localization (correlation length $R_0 \lesssim 0.12$ Å) ⇒ confronts the XENONnT/Majorana X-ray bounds ⇒ Z-Spin decoherence constrained or excluded.	**OBSERVATIONAL (existing data)**

Gate	Condition	Type
F-Q1-5th	An unscreened ε-mediated fifth force of strength ~ A = 0.08 at a laboratory–planetary range is excluded by Eöt-Wash/MICROSCOPE ⇒ ε must be heavy/short-range or screened; failure to satisfy this ⇒ ZS-F1 ε-sector constrained.	**OBSERVATIONAL (existing data)**

§10. Nine-Step Verification Protocol

Step	Check	Result
1	Zero free parameters / anti-numerology	PASS — only **A**, **Q**, dim(**Z**) and standard constants; Bennett/Nielsen–Chuang/Lawvere/Ishizaka–Hiroshima/Christandl carry no tunable constant; reach bound is a derived scale; No-Go and gate closures are structural (anti-numerology N/A).
2	Algebraic / cross-paper dependency	PASS — chain ZS-F0(Lawvere,*z**) → ZS-M1(*z** unique) → ZS-F1(L_XY=0, action) → ZS-F5(dim Z=2) → ZS-Q1(τ_D,Γ,σ_z) → ZS-Q2(τ_ent,area law) → ZS-Q7(ln 2) → ZS-Q16(linear ensemble, strong OPEN); no version conflict; *z** and **A** inherited unmodified.
3	Consistency with observation	PASS — reach bound contradicts no datum; no-programming/PBT/no-cloning are established QI; F-Q1-rad shows Z-Spin escapes the X-ray bounds; photonic teleportation lies in the cosmological-reach regime.
4	Epistemic status legend adequacy	PASS — DERIVED (reach bound, No-Go, §5.5) / TESTABLE (1/A) / DERIVED-interpretation (unification) / NON-CLAIM (
5	Multi-layer falsification gates	PASS — external (F-Q17.1/6/7), mathematical (F-Q17.2), observational (F-Q17.3/4, F-Q1-rad, F-Q1-5th), promotion (F-Q17.5).
6	Reference format (APS/arXiv)	PASS — see References.
7	Structure compliance (§0–Version History)	PASS.
8	Formatting guidelines	PASS — TNR 11pt base; headings 16/13/12pt bold; tables 9pt, 0.75pt borders, #F3F3F3 header (typeset edition).

Step	Check	Result
9	Typo / self-reference review	PASS — second-pass review completed; two prior over-claims corrected (v1 radiation; v2 Yukawa category error), reflected in §7 and Version History; no prior numerical claim retracted.

Companion scripts: zs_q17_v2_reach_bound_verify.py (14/14 PASS, reproduces both ZS-Q1 and ZS-Q2 anchors and the m^(−5/3) scaling), and zs_q1_radiation_gate_F-Q1-rad.py (the F-Q1-rad deciding calculation, structural + parametric).

§11. Conclusion

The v1.x question answered itself in the textbook — faithful but sub-luminal, the super-luminal version blocked by the framework's own no-signaling — so we demoted it to Tier-1 consistency and asked what is Z-Spin-specific. The **Reach Bound** L_max(m) = c·ℏ/(2A·E_diff) is a parameter-free, m^(−5/3) ceiling on coherent transport, reproducing the heritage anchors and carrying the falsifiable 1/A = 12.49 spatial signature into the near-term nanosphere and GIE programs; its strong-gravity limit gives E_diff = (3/10)mc^2 at the horizon, the rest-energy-set divergence of the X→Y ferrying that answers the singularity question and meets ZS-A3 at the horizon-as-Z-boundary. The **Self-Mediation No-Go** is the structural heart — the rank-2 mediator can route the matter and waves it mediates but cannot route itself — proven by no-programming and the Lawvere diagonal with z* unique, and now *completely formalized* by port-based teleportation: self-mediation is impossible deterministically and exactly, and approximately is fidelity-ceilinged by F_2(n), reaching unity only in the $n \to \infty$ i-tetration limit. This single obstruction, read at three layers, is simultaneously the bootstrap B0, the strong-outcome residue, and self-mediation. Finally, confronting current data, we close **F-Q1-rad**: Z-Spin is a scalar-tensor / environmental model with no universal radiating term and a Yukawa ε that decouples when heavy, so it escapes the X-ray bounds that killed parameter-free Diósi–Penrose; the same gating makes the discriminating test interferometric (2028–2032), and a separate fifth-force gate **F-Q1-5th** is registered for the ε-sector. The corpus gains a completely-formalized no-go theorem, a falsifiable transport scale, and two closed/registered experimental gates, in place of a relabelled textbook channel. The one residual OPEN — a dynamical single-outcome rule, which would promote the strong-outcome face — is the same diagonal at the single-run layer, left honestly unclosed; and the one residual quantitative gap — V(ε), hence m_ε — does not affect the structural closures and is the natural next sub-calculation.

Acknowledgements & Code Availability

This paper consolidates internal Z-Spin Collaboration research notes and imports external proven mathematics as cited. Reach-bound predictions are verified by zs_q17_v2_reach_bound_verify.py (14/14 PASS); the F-Q1-rad analysis is documented in zs_q1_radiation_gate_F-Q1-rad.py. All decoherence anchors (τ_D, τ_ent, M_crit) are inherited from the verified suites of ZS-Q1 and ZS-Q2. The No-Go and the gate closures are structural theorems/arguments and are not asserted to be numerically "verified" by code.

Appendix A — Reach-Bound Derivation Detail. From $\Gamma = 2A(\Delta E/\hbar)^2$ (ZS-Q1 §3.4) the single-system off-diagonal decays as $\exp(-\Gamma t)$; for a Bell pair both halves dephase, so concurrence $C(t) = \exp(-2\Gamma t)$ and $\tau_{ent} = \tau_{single}/2$ (ZS-Q2 §7.3). Resource survival to record arrival requires $\tau_{ent} \geq L/c$, i.e. $L \leq c\cdot\tau_{ent} = L_{max}$. With $E_{diff} = (3/5)G\, m^2/R$ and $R = (3m/4\pi\rho)^{(1/3)}$, $L_{max} \propto m^{(-5/3)}$. The Penrose comparison uses $A \to 1$, giving the mass-independent ratio $1/A$.

Appendix B — The No-Programming Theorem in Z-Spin Language. Nielsen–Chuang (1997): a deterministic programmable array storing *N* inequivalent unitaries needs orthogonal program states, dim(program) $\geq N$. The only cross-sector conduit is the rank-2 Z-bottleneck ($L_{XY} \equiv 0$; capacity $\leq \ln 2$). **Λ_Z** spans $\gg 2$ inequivalent operations on **Q** = 11, so it cannot be programmed through a dim-2 conduit; the single external qubit is the $N = 2$ permitted case. Pati–Braunstein no-deleting (2000) is a corollary of the same orthogonality.

Appendix C — The Lawvere Diagonal and *z Uniqueness.** Lawvere (1969): point-surjectivity $A \to Y^A$ forces a fixed point for every endomap of *Y* (the categorical Cantor/Gödel skeleton, used in ZS-F0 §2.2, §11). The Z-Spin self-application is $T(z) = i^z$, unique attractor z^* (ZS-M1, Lambert W $k = 0$), the collapse fixed point; a self-transport would be a second such fixed point, forbidden.

Appendix D — The Horizon Corollary and the ZS-A3 Connection. At $R_s = 2Gm/c^2$, Eq. (3) gives $E_{diff} = (3/10)mc^2$, so $\tau_H = 10\hbar/(3Amc^2)$ and $\Gamma_H \propto m^2$. ZS-A3 v1.0: inside the horizon *r* is timelike and *t* spacelike (an $r\leftrightarrow t$ exchange identified with X↔Y), the horizon is the Z-sector boundary ($\varepsilon = 0$), a black hole an X-structure collapsed into a Y-prison. This appendix asserts only the Newtonian $(3/10)mc^2$ identity (DERIVED) and the inherited $r\leftrightarrow t$ = X↔Y reading (HYPOTHESIS).

Appendix E — Port-Based Teleportation and the Complete Formalization [NEW v2.1]. PBT (Ishizaka–Hiroshima 2008) realizes an approximate (dPBT) or probabilistic-exact (pPBT) universal programmable processor with finite resources, faithful only as $N \to \infty$ (forced by the no-programming theorem). The deterministic entanglement fidelity obeys $F_d(N) \leq \sqrt{N}/d$ ($N \leq d^2/2$), $1 - (d^2-1)/(16N^2)$ otherwise (Christandl *et al.* 2021), with closed forms for *d* and small *N* (Studziński *et al.*). For $d = 2$ and *N* serial Z-handshakes, self-mediation fidelity is ceilinged below unity for finite *N*, reaching unity only as $N \to \infty$; identifying *N* with the handshake count *n* (ZS-F10), this $N \to \infty$ asymptote is the i-tetration convergence to *z** (ZS-F0/M1). Self-mediation is impossible deterministically and exactly; approximately it is bounded by $F_2(n)$.

Appendix F — The F-Q1-rad Deciding Calculation [NEW v2.1]. Radiation rate $\propto$ momentum diffusion D_{pp} (Bassi–Donadi). The X-ray experiments bound the stochastic-collapse class (universal white noise; smaller smearing ⇒ more radiation; $R_0 \to 0$ excluded). ZS-Q1 is scalar-tensor/environmental: ε is a deterministic massive field, decoherence $\propto E_{diff}$ (gated, = 0 unsuperposed), ensemble linear (ZS-Q16) — no universal noise. ε is a Yukawa mediator: heavy ⇒ decouples ($e^{(-r/\lambda_\varepsilon)}$), opposite to the DP $1/R_0^3$ enhancement; the residual ε-vacuum effect is $(A\cdot G)^2$-suppressed second order with the ε mass as natural cutoff. The naive "λ_ε as smearing length" worry is a category error. Hence F-Q1-rad closes for all m_ε. A separate fifth-force gate F-Q1-5th follows from the A-strength Yukawa, requiring heavy/screened ε.

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[20] Z-Spin Cosmology (2026): ZS-F0 (bootstrap; Lawvere; FFPP), ZS-F1 (action; L_XY ≡ 0; U(1) Z-bias Φ), ZS-F2 (**A** = 35/437), ZS-F5 (**Q** = 11; dim Z = 2), ZS-M1 (HSI; *z**), ZS-Q1 (geometric decoherence; τ_D), ZS-Q2 (entanglement; τ_ent; area law), ZS-Q7 (Z-bottleneck; ≤ ln 2), ZS-Q12 (self-referential collapse; *z**), ZS-Q16 (outcome problem; linear ensemble; strong OPEN), ZS-T1

(information routing; serial handshakes), ZS-A3 (black-hole ε-field; horizon = Z-boundary), ZS-F18 (meta-map).

Version History

v2.1 (June 2026): Adds the **complete formalization of the Self-Mediation No-Go** via port-based teleportation (§5.5, Appendix E): deterministic-exact PROVEN, approximate fidelity ceiling $F_d(N) \leq \sqrt{N}/d$ ($d = 2$) with the port count identified as the serial handshake count and unit fidelity only in the $n \to \infty$ (i-tetration z^*) limit. Adds **§7 (experimental status and gates)**: the 2026 interferometry/GIE landscape for the ratio-12.49 test; the deciding analysis closing **F-Q1-rad** (ZS-Q1 is scalar-tensor/environmental with no universal radiating term, and ε is a Yukawa mediator that decouples when heavy — escaping the X-ray bounds that excluded parameter-free Diósi–Penrose); and the new **F-Q1-5th** fifth-force gate (Appendix F). Honest corrections incorporated: the v1-era radiation over-claim (a dimensionally-suspect quadratic rate) and a v2-era category error (ε Compton wavelength mistaken for a Diósi–Penrose smearing length) are corrected and reflected in §7; no prior numerical claim retracted. New gates F-Q17.7, F-Q1-rad, F-Q1-5th. New companion script zs_q1_radiation_gate_F-Q1-rad.py. (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; zero free parameters.

v2.0 (June 2026): Major reorientation. Demoted the v1.x faithful-sub-luminal-teleportation [DERIVED] and FTL [NON-CLAIM] to Tier-1 consistency (§8). Introduced the **Reach Bound** [DERIVED/TESTABLE] with $m^{-5/3}$ scaling, both heritage anchors, the 1/A = 12.49 spatial signature, and the Horizon Corollary (§3); and the **Self-Mediation No-Go** [DERIVED] via no-programming + Lawvere/z^* (§5), unified with the ZS-F0 bootstrap B0 and the ZS-Q16 strong-outcome OPEN (§6). Companion script zs_q17_v2_reach_bound_verify.py (14/14 PASS).

v1.2 / v1.1 / v1.0 (June 2026): [SUPERSEDED] Teleportation framing; faithful sub-luminal [DERIVED], FTL [NON-CLAIM]; inter-cell L_XY closure; dual-frame proper-time-compression reading. The dual-frame §6.2–6.3 content is relocated to ZS-F19; the observer/outcome content is subsumed by ZS-Q16 v2.5. Retained only as Tier-1 consistency (§8).